



Caringbah High School

2018

Trial HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen only
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I 10 marks

Pages 2 – 6

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II 90 marks

Pages 7 – 15

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Question 1 - 10 (1 mark each) Answer on page provided.

1 Let $z = 4 - 3i$. What is the value of $\overline{iz}$?

A) $3 + 4i$

B) $3 - 4i$

C) $-3 + 4i$

D) $-3 - 4i$

2 What is the remainder when $x^3 + x^2 + 5x + 6$ is divided by $(x + i)$?

A) $7 - 4i$

B) $7 - 6i$

C) $5 - 4i$

D) $5 + 6i$

3 The roots of the equation $2x^3 - 3x^2 + 2x + 2 = 0$ are α, β and γ .

What is the value of $\alpha + \beta - \frac{1}{\alpha\beta}$?

A) -1

B) 1

C) $-\frac{3}{2}$

D) $\frac{3}{2}$

4 What are the coordinates of the foci of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$?

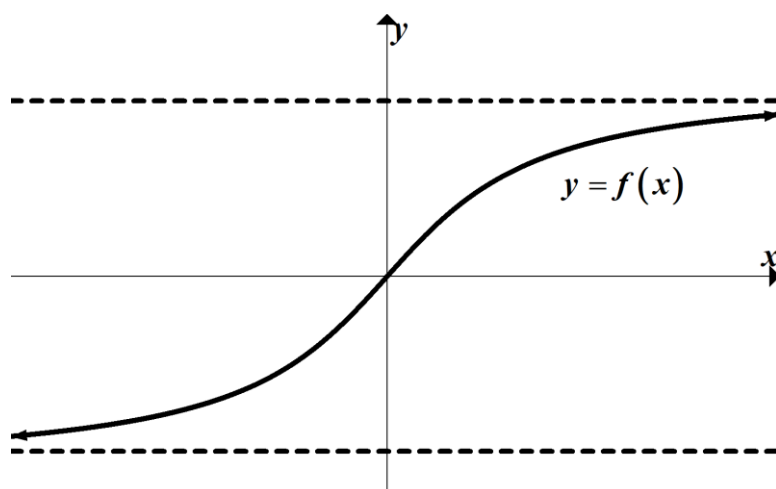
A) $\left(\pm \frac{2\sqrt{5}}{3}, 0\right)$

B) $\left(0, \pm \frac{2\sqrt{5}}{3}\right)$

C) $(\pm\sqrt{5}, 0)$

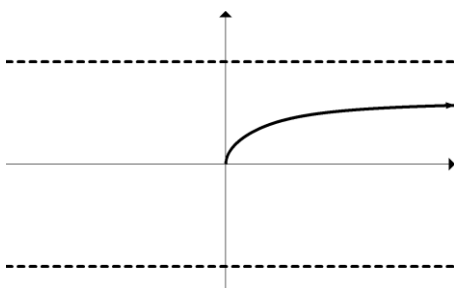
D) $(0, \pm\sqrt{5})$

- 5 The graph of $y = f(x)$ is shown below?

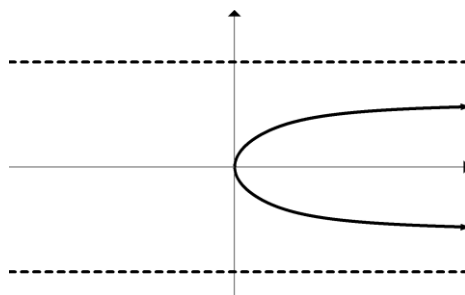


Which of the following graphs best represents $y^2 = f(x)$?

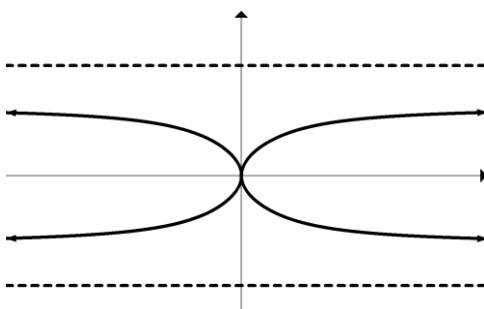
A)



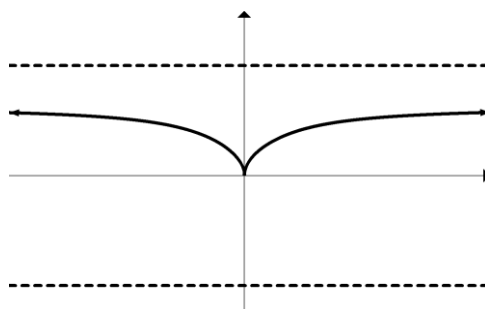
B)



C)



D)



6 Consider $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 x}{(1 + \tan x)^2} dx$.

After using an appropriate substitution, which of the following is equivalent to I ?

A) $\int_0^2 \frac{1}{u^2} du$

B) $\int_0^2 \frac{u^2}{(1+u)^3} du$

C) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u} du$

D) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u^3} du$

7 If $e^x + e^y = 1$, which of the following is an expression for $\frac{dy}{dx}$?

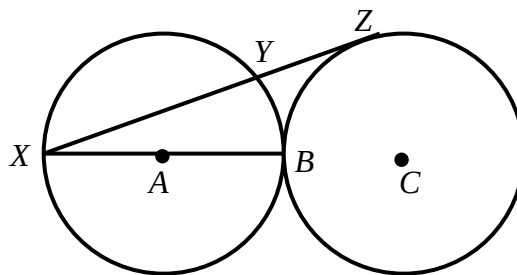
A) $-e^{x-y}$

B) $-e^{y-x}$

C) e^{x-y}

D) e^{y-x}

- 8 Two equal circles touch externally at B . XB is a diameter of the circle centred at A . XZ is the tangent from X to the circle centred at C and cuts the first circle at Y .



Which is the correct expression that relates XZ to XY ?

A) $3XZ = 4XY$

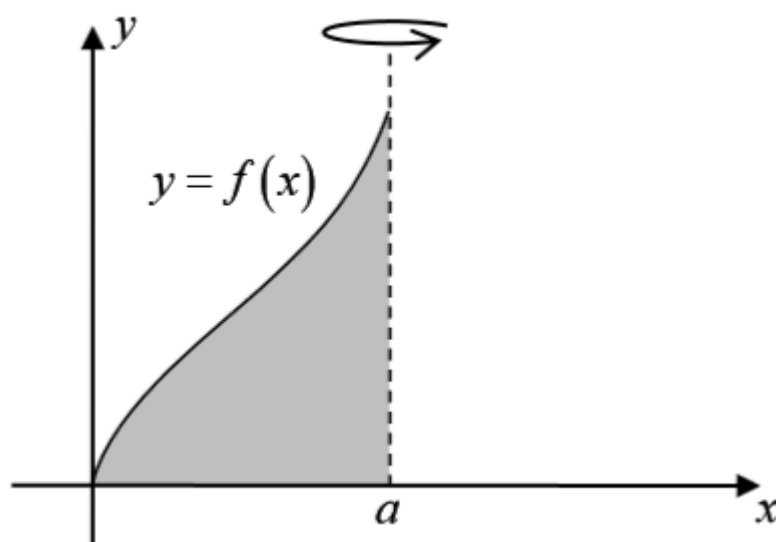
B) $XZ = 2XY$

C) $2XZ = 3XY$

D) $2XZ = 5XY$

- 9 The function $y = f(x)$ is monotonic increasing over the interval $0 \leq x \leq a$.

The region bounded by this function, the x -axis and the line $x = a$ is to be rotated about the line $x = a$ to form a solid of revolution.



Which of the following integrals represents the volume of this solid?

- A) $\pi \int_0^a [a - f(x)]^2 dx$ B) $\pi \int_0^a [a - x]^2 dx$
- C) $\pi \int_0^{f(a)} [a - f^{-1}(y)]^2 dy$ D) $\pi \int_0^{f(a)} [a - y]^2 dy$

- 10 Consider $\omega = \frac{1}{2}(-1 + i\sqrt{3})$.

Which of the following is a correct expression for $|z|$, where $z = a + b\omega$ and a and b are constants.

A) $\sqrt{(a - b)^2 + 2ab}$

B) $\sqrt{(a - b)^2 + ab}$

C) $\sqrt{(a - b)^2 - ab}$

D) $\sqrt{(a - b)^2 - 2ab}$

END OF MULTIPLE CHOICE QUESTIONS

Section II**90 marks****Attempt all questions 11–16****Allow about 2 hours and 45 minutes for this section**

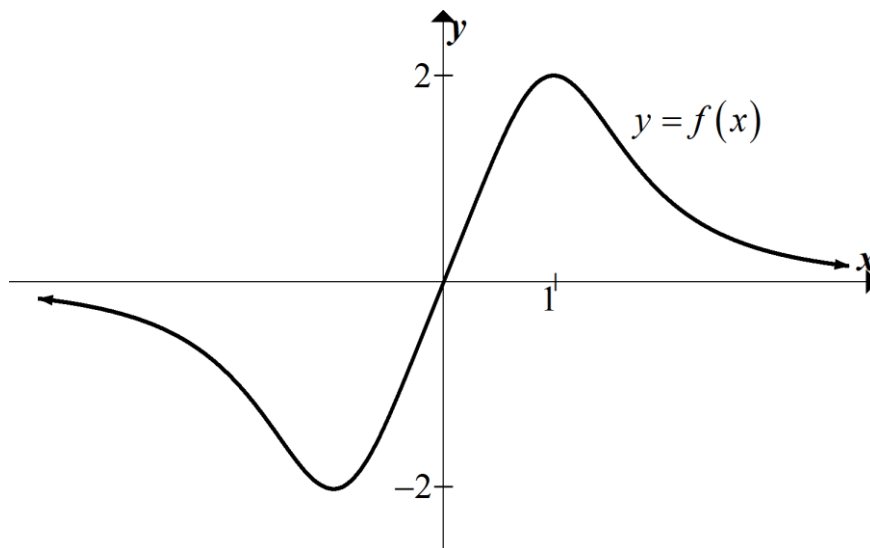
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.	Marks
a) Let $z = 1 - i$ and $u = 3 + 2i$. Find:	
(i) $\operatorname{Im}(u - z)$.	1
(ii) $z\bar{u}$.	1
b) Let the point P represent the complex number z in the Argand diagram.	1
Describe the features of the new point Q representing the complex number $2iz$.	
c) Find $\int \frac{dx}{x^2 + 6x + 13}$.	2
d) It is given that $Z = 1 - \sqrt{3}i$.	
(i) Express Z in modulus – argument form.	2
(ii) Hence, express Z^{10} in $x + iy$ form.	2
e) Find the x -coordinates of the points on the curve $2x^2 + 2xy + 3y^2 = 15$ where the tangents to the curve are vertical.	3
f) (i) Find the domain and range of $f(x) = \tan^{-1}(e^x)$.	2
(ii) Sketch $y = f(x)$ showing any intercepts and asymptotes.	1

Question 12 (15 marks) Start a NEW booklet.**Marks**

- a) The diagram below shows the graph of $y = f(x)$ which is an odd function.



On separate sketches, showing any important features, neatly draw the graphs of

- | | | |
|-------|----------------------|----------|
| (i) | $y = f(-x)$ | 1 |
| (ii) | $y = \sqrt{f(x)}$ | 2 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv) | $y = x + f(x)$ | 2 |
| (v) | $y = f'(x)$ | 2 |

Question 12 continues on page 9

Question 12 (continued)

Marks

- b) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate $\int_0^{\frac{\pi}{3}} \frac{d\theta}{1 + \sin \theta}$. 3
- c) C is a unit circle with its centre at the origin on the Argand diagram. If the point representing z_1 moves on the circle C and $z_2 = \frac{1-i}{z_1}$, find the locus of z_2 and describe it geometrically. 3

End of Question 12

Question 13 (15 marks) Start a NEW booklet.**Marks**

- a) Consider the polynomial $P(x) = x^4 - 8x^3 + 18x^2 - 27$. 3

Factorise $P(x)$, given that it has a root of multiplicity 3.

- b) Find $\int \frac{1}{x^2} \sqrt{1 + \frac{4}{x}} dx$. 3

- c) Express $\frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta + i \sin \theta}$ in the form $a + ib$. 3

- d) The roots of the cubic equation $x^3 - 4x^2 - 12 = 0$ are α, β and γ .

Find the equation with roots:

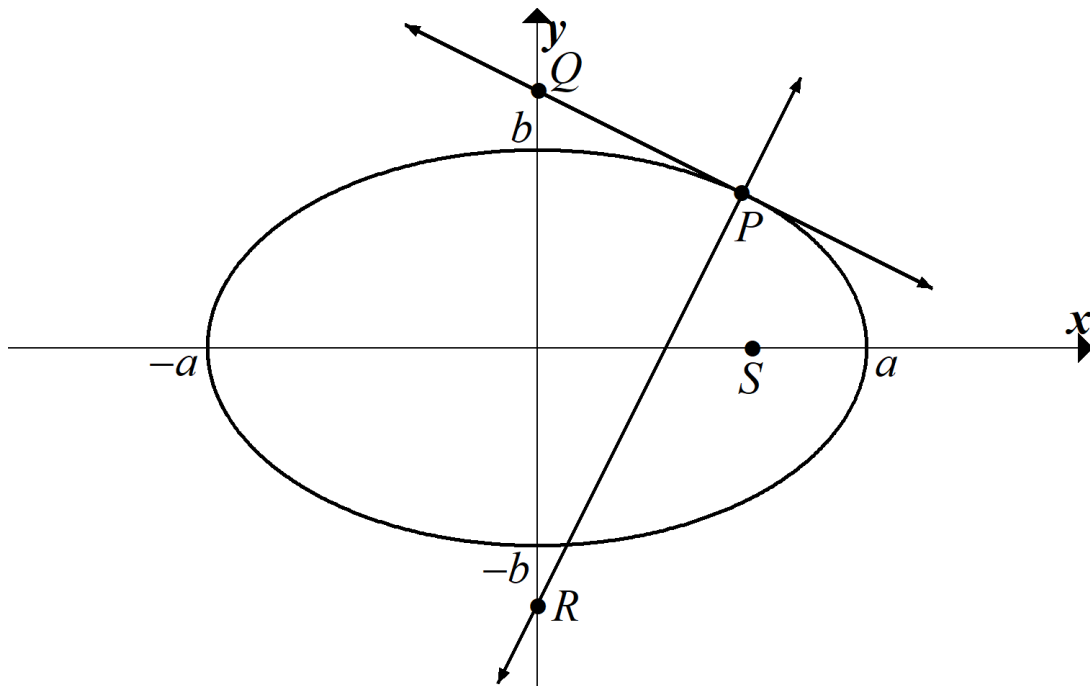
- (i) $-\alpha, -\beta, -\gamma$. 2

- (ii) $\alpha + \beta, \beta + \gamma, \gamma + \alpha$. 2

- e) Solve $a^x = e^{2x-1}$ for x , expressing the answer in terms of a . ($a > 0$). 2

Question 14 (15 marks) Start a NEW booklet.**Marks**

- a) The point $P(a\cos\theta, b\sin\theta)$ lies on the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ as shown below.



The tangent at P cuts the y -axis at Q . Also, the normal at P cuts the y -axis at R .

The normal at P is: $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$. [Do not prove this]

- (i) Show that the equation of the tangent at P is given by 2

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1.$$

- (ii) Show that the y -coordinate of Q is $\frac{b}{\sin\theta}$. 1

- (iii) Write down the coordinates of the point R . 1

- (iv) If S is the focus $(ae, 0)$, prove that P, Q, R and S are concyclic points. 3

Question 14 continues on page 12

Question 14 (continued)

Marks

- b) Consider the function $f(x) = e^{-x} - 1 + x$.

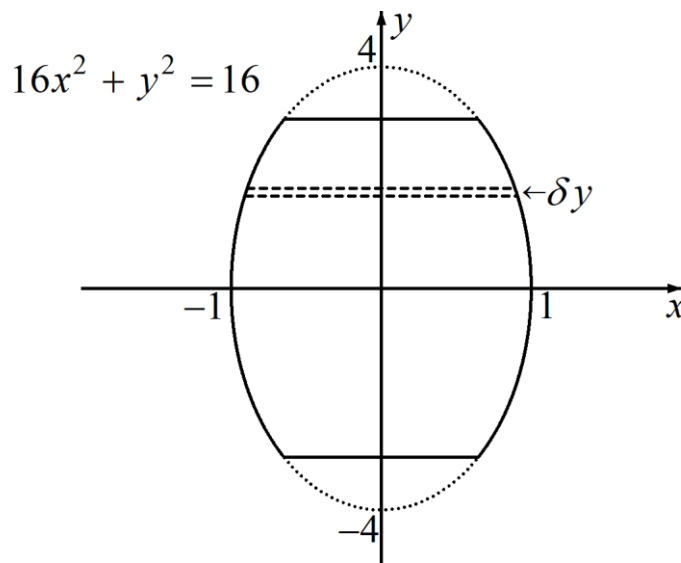
By finding expressions for $f'(x)$ and $f''(x)$ show that

3

$$e^{-x} \geq 1 - x, \text{ for all } x.$$

- c) At a children's playground there is a narrow 6 metre long tunnel with vertical walls. The base (or floor) of the tunnel is in the shape of an ellipse with equal 1 metre amounts cut from each end.

Vertical cross-sections perpendicular to the major axis of the ellipse are in the shape of a rectangle topped by a semi-circle. The base of the rectangle is twice its height and the ellipse has equation $16x^2 + y^2 = 16$.



- (i) With the aid of a diagram show that the area of a typical cross-section

2

is given by $A_s = \left(\frac{4 + \pi}{2} \right) x^2$.

- (ii) Hence, find the volume of the tunnel in exact form.

3

End of Question 14

Question 15 (15 marks) Start a NEW booklet.**Marks**

a) Find the general solution in radians to the equation $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}$. **3**

b) (i) Let x be a fixed, non-zero number satisfying $x > -1$. **3**

Use the method of mathematical induction to prove that for $n = 2, 3, 4, \dots$

$$(1 + x)^n > 1 + nx$$

(ii) Deduce that $\left(1 - \frac{1}{2n}\right)^n > \frac{1}{2}$ for $n = 2, 3, 4, \dots$ **1**

c) If $a > 0$ and $b > 0$ show that $a^3 + b^3 \geq a^2b + ab^2$. [Do not use Induction] **2**

d) $P\left(5p, \frac{5}{p}\right), p > 0$ and $Q\left(5q, \frac{5}{q}\right), q > 0$ are two points on the hyperbola $xy = 25$.

The equation of the chord PQ is given by $x + pqy = 5(p + q)$. [Do not prove this]

(i) Hence or otherwise, state the equations of the tangents at P and Q . **1**

(ii) The tangents at P and Q intersect at R . **2**

Show that the coordinates of R are given by $R\left(\frac{10pq}{p+q}, \frac{10}{p+q}\right)$.

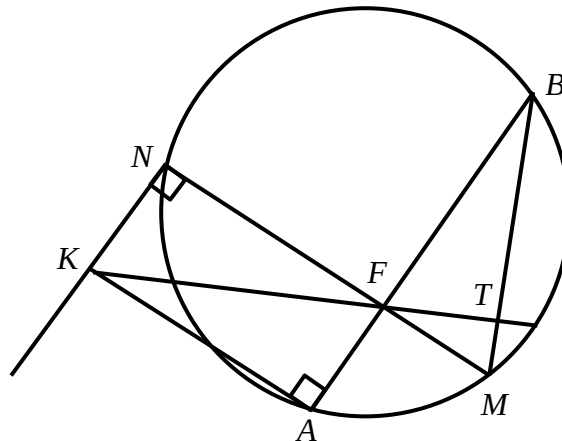
(iii) If the secant PQ passes through the point $S(15, 0)$, find the locus of R . **3**

Question 16 (15 marks) Start a NEW booklet.**Marks**

- a) The area enclosed between the curves $y = \sqrt{x}$ and $y = x^2$ is rotated about the y -axis through one complete revolution. Use the method of cylindrical shells to find the volume of the solid that is generated. **3**

- b) If $I_n = \int_0^1 x^n e^{-x} dx$, show that $I_n = -\frac{1}{e} + n I_{n-1}$. **3**

- c) As shown below, a circle has two chords AB and MN intersecting at F . Perpendiculars are drawn to these chords at A and N intersecting at K . KF produced meets MB at T .



Answer this question on the page provided

- (i) Explain why $AKNF$ is a cyclic quadrilateral. **1**
- (ii) By letting $\angle AKF = \theta$, prove that KT is perpendicular to MB . **3**

[Use and hand in the attached diagram - see page 18]

Question 16 continues on page 15

Question 16 (continued)

Marks

- d) Water flows into an aquarium at a rate proportional to the amount of water Q in the aquarium. At the same time, evaporation occurs at a rate proportional to the square of the quantity of water in the aquarium. Thus at any time ' t ', it is known that

$$\frac{dQ}{dt} = aQ - bQ^2$$

where ' a ' and ' b ' are constants. Initially $Q = Q_0$.

- i) Using partial fractions, show that an expression for Q in terms of t

4

is given by $Q = \frac{aQ_0 e^{at}}{(a - bQ_0) + bQ_0 e^{at}}.$

- ii) Show that the quantity of water tends to a limit as time increases.

1

END OF EXAM

Candidate Name/Number: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
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6.	
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10.	

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CHS YEAR 12 EXTENSION 2 2018	TRIAL HSC SOLUTIONS
Multiple Choice Section: 1.B 2.C 3.D 4.D 5.B 6.A 7.A 8.C 9.C 10.B Question 1. $iz = i(4-3i)$ $iz = 3 + 4i$ $\therefore \bar{iz} = 3 - 4i \quad \text{-----} \boxed{B}$ Question 2. The remainder is given by $P(i) = (-i)^3 + (-i)^2 + 5(-i) + 6$ $= i - 1 - 5i + 6$ $= 5 - 4i \quad \text{-----} \boxed{C}$ Question 3. $\alpha + \beta + \gamma = \frac{3}{2}$ $\alpha \beta \gamma = -1 \rightarrow \gamma = -\frac{1}{\alpha\beta}$ $\therefore \alpha + \beta - \frac{1}{\alpha\beta} = \frac{3}{2} \quad \text{-----} \boxed{D}$ Question 4. $a=2; b=3; a^2 = b^2(1-e^2); \text{Foci } (0, \pm be)$ $\therefore e^2 = \frac{5}{9} \rightarrow e = \frac{\sqrt{5}}{3}$ $\therefore \text{Foci } (0, \pm\sqrt{5}) \quad \text{-----} \boxed{D}$ Question 5. ----- $\boxed{B}$	Question 6. let $u = 1 + \tan x \rightarrow du = \sec^2 x \, dx$ when $x = \frac{\pi}{4}, u = 2; x = -\frac{\pi}{4}, u = 0.$ $\therefore I = \int_0^2 \frac{1}{u^2} du \quad \text{-----} \boxed{A}$ Question 7. $e^x + e^y = 1$ and using implicit differentiation $e^x + e^y \frac{dy}{dx} = 0$ $\therefore \frac{dy}{dx} = -\frac{e^x}{e^y} = -e^{x-y} \quad \text{-----} \boxed{A}$ Question 8. $\angle XYB = 90^\circ$ and $\angle XZXC = 90^\circ.$ $\therefore \frac{XZ}{XY} = \frac{XC}{XB} = \frac{3ZC}{2ZC} = \frac{3}{2}$ $\therefore 2XZ = 3XY \quad \text{-----} \boxed{C}$ Question 9. The radius of a typical slice is $r = a - x$ $\therefore V_s = \pi r^2 \delta y$ Volume $= \pi \int_0^{f(a)} (a-x)^2 dy$ $= \pi \int_0^{f(a)} [a - f^{-1}(y)]^2 dy \quad \text{-----} \boxed{C}$

Question 10.

$$|z| = |a + b\omega|$$

$$= \left| a + b \left(\frac{1}{2}(-1 + i\sqrt{3}) \right) \right|$$

$$= \left| \left(a - \frac{b}{2} \right) + \frac{b\sqrt{3}}{2}i \right|$$

$$= \sqrt{\left(a - \frac{b}{2} \right)^2 + \left(\frac{b\sqrt{3}}{2} \right)^2}$$

$$= \sqrt{a^2 - ab + \frac{b^2}{4} + \frac{3b^2}{4}}$$

$$= \sqrt{a^2 - 2ab + b^2 + ab}$$

$$= \sqrt{(a-b)^2 + ab} \quad \text{-----} \boxed{B}$$

Question 11

a) i) $\text{Im}(u - z) = \text{Im}(2 + 3i) = 3$

ii) $z\bar{u} = (1 - i)(3 - 2i) = 1 - 5i$

b) It is rotated 90° in an anti-clockwise direction
and $|Q| = 2|P|$.

$$\begin{aligned} \text{c) } \int \frac{dx}{x^2 + 6x + 13} &= \int \frac{dx}{(x^2 + 6x + 9) + 4} \\ &= \int \frac{dx}{(x+3)^2 + 2^2} \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + c \end{aligned}$$

d) i) $Z = 1 - \sqrt{3}i$

$$\therefore \arg(Z) = -\frac{\pi}{3} \text{ and } |Z| = 2$$

$$\therefore Z = 2\text{cis}\left(-\frac{\pi}{3}\right)$$

d) ii) $Z^{10} = \left[2\text{cis}\left(-\frac{\pi}{3}\right) \right]^{10}$

$$= 2^{10} \text{cis}\left(-\frac{10\pi}{3}\right)$$

$$= 1024 \left[\cos\left(\frac{10\pi}{3}\right) - i \sin\left(\frac{10\pi}{3}\right) \right]$$

$$= 1024 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$= 512(-1 + \sqrt{3}i)$$

e) $2x^2 + 2xy + 3y^2 = 15$

Using implicit differentiation

$$4x + 2x \cdot y' + 2y + 6y \cdot y' = 0$$

$$2x + x \cdot y' + y + 3y \cdot y' = 0$$

$$y'(x + 3y) = -2x - y$$

$$\therefore \frac{dy}{dx} = \frac{-2x - y}{x + 3y}$$

vertical tangents will occur when $x + 3y = 0$

i.e. when $y = -\frac{x}{3}$

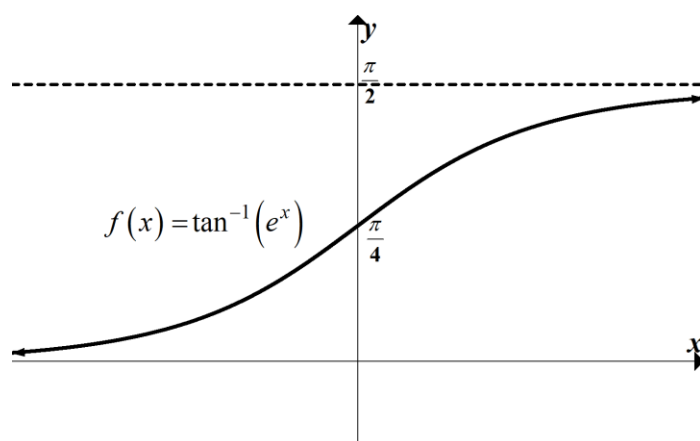
$$\therefore 2x^2 + 2x\left(-\frac{x}{3}\right) + 3\left(-\frac{x}{3}\right)^2 = 15$$

$$\therefore x^2 = 9 \rightarrow x = \pm 3$$

f) i) D: all real x .

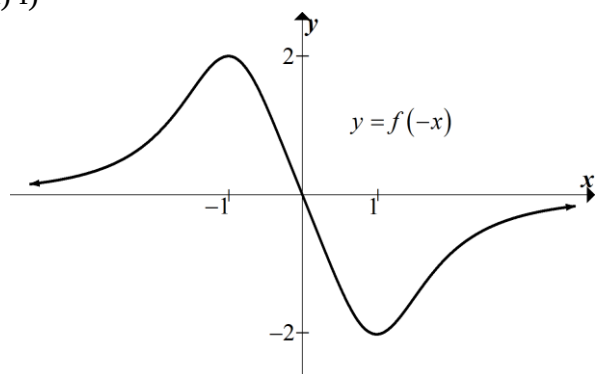
$$\text{R: } 0 < y < \frac{\pi}{2}.$$

ii)

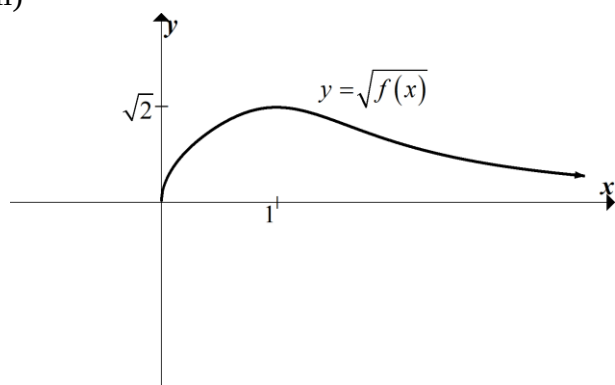


Question 12.

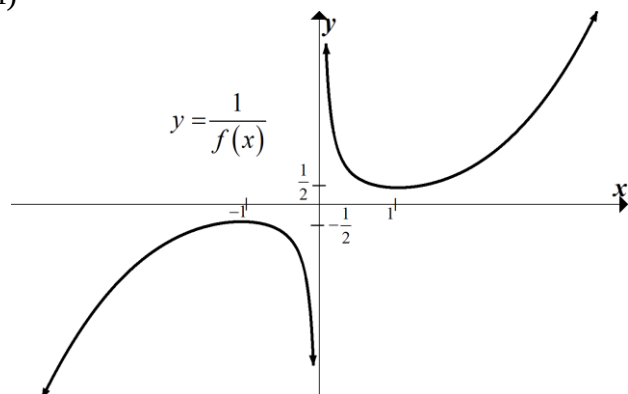
a) i)



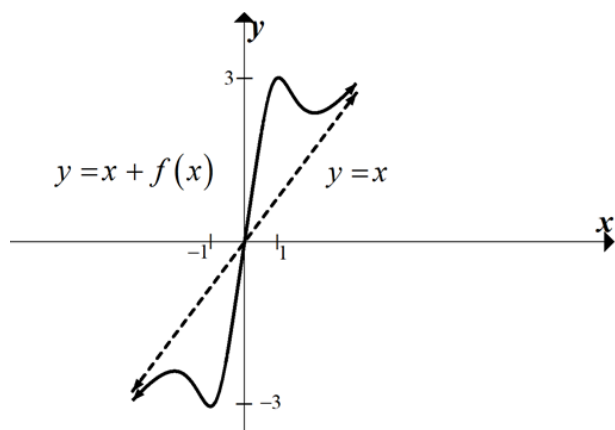
ii)



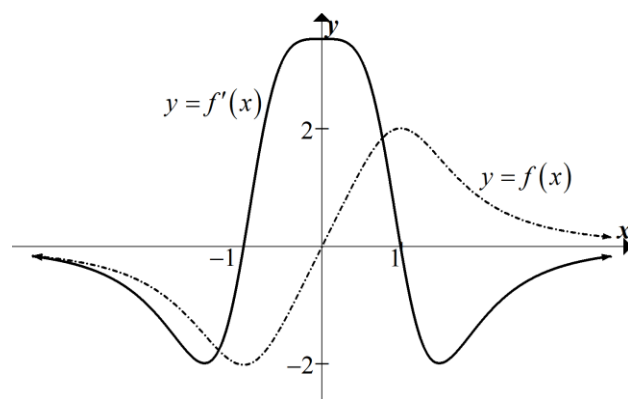
iii)



iv)



v)



$$b) \quad t = \tan \frac{\theta}{2} \rightarrow d\theta = \frac{2}{1+t^2} dt$$

$$\text{When } \theta = 0, t = 0; \theta = \frac{\pi}{3}, t = \frac{1}{\sqrt{3}}. \text{ Also } \sin \theta = \frac{2t}{1+t^2}$$

$$I = \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2t}{1+t^2} dt$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1+2t+t^2} dt = 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{(1+t)^2} dt$$

$$= -2 \left[\frac{1}{1+t} \right]_0^{\frac{1}{\sqrt{3}}} = \frac{2}{1+\sqrt{3}}$$

$$c) \quad z_2 = \frac{1-i}{z_1} \rightarrow |z_2| = \left| \frac{1-i}{z_1} \right|$$

$$\therefore |z_2| = \frac{|1-i|}{|z_1|} = \frac{\sqrt{2}}{1} \text{ as } z_1 \text{ is on the unit circle.}$$

$$\therefore |z_2| = \sqrt{2}$$

and hence the locus of z_2 is a circle centre (0,0) and radius $\sqrt{2}$.

Question 13.

a) $P(x) = x^4 - 8x^3 + 18x^2 - 27.$

$$P'(x) = 4x^3 - 24x^2 + 36x$$

$$P''(x) = 12x^2 - 48x + 36$$

For a triple root $P''(x) = 0$

$$\therefore 12x^2 - 48x + 36 = 0$$

$$\therefore 12(x-3)(x-1) = 0$$

$$\therefore x = 3 \text{ or } x = 1$$

also since $P(3) = 3^4 - 8 \times 3^3 + 18 \times 3^2 - 27 = 0$

then $x = 3$ is the root of multiplicity three.

[Note $P(1) = 1 - 8 + 18 - 27 \neq 0$]

$$\therefore P(x) = (x-3)^3(x+1)$$

b) Let $I = \int \frac{1}{x^2} \sqrt{1 + \frac{4}{x}} dx$

$$\text{Let } u = \frac{1}{x} \rightarrow du = -\frac{1}{x^2} dx$$

$$\therefore I = -\int \sqrt{1+4u} du$$

$$= -\int (1+4u)^{1/2} du$$

$$= -\frac{(1+4u)^{3/2}}{4} \times \frac{2}{3}$$

$$= -\frac{(1+4u)^{3/2}}{6}$$

$$= -\frac{1}{6} \left(1 + \frac{4}{x}\right)^{3/2} + c$$

$$c) = \frac{(1 + \cos \theta - i \sin \theta)}{(1 + \cos \theta + i \sin \theta)} \times \frac{(1 + \cos \theta - i \sin \theta)}{(1 + \cos \theta - i \sin \theta)}$$

$$= \frac{(1 + \cos \theta - i \sin \theta)^2}{(1 + \cos \theta)^2 - (i \sin \theta)^2}$$

$$= \frac{(1 + \cos \theta)^2 - 2(1 + \cos \theta)i \sin \theta + (i \sin \theta)^2}{(1 + \cos \theta)^2 - (i \sin \theta)^2}$$

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta - 2(1 + \cos \theta)i \sin \theta - \sin^2 \theta}{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}$$

$$= \frac{(\cos^2 \theta + \sin^2 \theta) + 2 \cos \theta + \cos^2 \theta - 2(1 + \cos \theta)i \sin \theta - \sin^2 \theta}{2 + 2 \cos \theta}$$

$$= \frac{2 \cos^2 \theta + 2 \cos \theta - 2(1 + \cos \theta)i \sin \theta}{2(1 + \cos \theta)}$$

$$= \frac{\cos \theta (1 + \cos \theta) - (1 + \cos \theta)i \sin \theta}{(1 + \cos \theta)}$$

$$= \cos \theta - i \sin \theta \text{ (see end for alternate solution)}$$

d) i) Let $x = -\alpha \rightarrow \alpha = -x$

$\therefore$ the required equation is given by

$$\therefore (-x)^3 - 4(-x)^2 - 12 = 0$$

$$\therefore -x^3 - 4x^2 - 12 = 0$$

$$\text{hence } x^3 + 4x^2 + 12 = 0$$

ii) The roots are of the form $\alpha + \beta + \gamma - \gamma$

$$= (\alpha + \beta + \gamma) - \gamma$$

$$= 4 - \gamma \text{ using sum of the roots.}$$

Hence let $x = 4 - \gamma \rightarrow \gamma = 4 - x$

$\therefore$ the required equation is given by

$$\therefore (4-x)^3 - 4(4-x)^2 - 12 = 0$$

$$\therefore 64 - 48x + 12x^2 - x^3 - 64 + 32x - 4x^2 - 12 = 0$$

$$\therefore x^3 - 8x^2 + 16x + 12 = 0$$

$$e) a^x = e^{2x-1}$$

$$\therefore \ln(a^x) = \ln(e^{2x-1})$$

$$\therefore x \ln a = 2x - 1$$

$$\therefore 1 = x(2 - \ln a)$$

$$\therefore x = \frac{1}{2 - \ln a}$$

Question 14.

a) i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by implicit differentiation

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0 \rightarrow \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\therefore \text{at } P(a \cos \theta, b \sin \theta), \frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$$

$$\therefore \text{eq}^n \text{ of } T \text{ at } P: y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$$

$$\text{Hence } \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \text{ ----- [1]}$$

ii) Using [1] and putting $x = 0$

$$\frac{y \sin \theta}{b} = 1 \rightarrow y = \frac{b}{\sin \theta}$$

iii) Using the equation of the normal [given] and putting $x = 0$

$$-\frac{by}{\sin \theta} = a^2 - b^2 \rightarrow y = \frac{b^2 - a^2}{b} \sin \theta$$

$$\therefore R\left(0, \frac{b^2 - a^2}{b} \sin \theta\right)$$

iii) $\angle QPR = 90^\circ$ ($\angle$ btwn tangent and normal)

$$m_{QS} = \frac{0 - \frac{b}{\sin \theta}}{ae - 0} = \frac{-b}{ae \sin \theta}$$

$$m_{RS} = \frac{\frac{b^2 - a^2}{b} \sin \theta - 0}{0 - ae} = \frac{\frac{b^2 - a^2}{b} \sin \theta}{-ae}$$

$$\text{Now } b^2 = a^2(1 - e^2) \rightarrow b^2 - a^2 = -a^2 e^2$$

$$\therefore m_{RS} = \frac{-a^2 e^2 \sin \theta}{-abe} = \frac{ae \sin \theta}{b}$$

$$\therefore m_{QS} \times m_{RS} = \frac{-b}{ae \sin \theta} \times \frac{ae \sin \theta}{b} = -1$$

$$\therefore \angle QSR = 90^\circ = \angle QPR$$

Hence $QPSR$ are concyclic points
[$\angle$'s in the same segment]

b) $f(x) = e^{-x} - 1 + x$

$$\therefore f'(x) = -e^{-x} + 1 \text{ and } f''(x) = e^{-x}$$

$$\text{As } f''(x) = e^{-x} > 0 \text{ for all } x$$

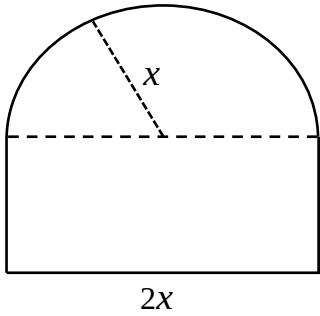
then $f(x)$ is concave up

$$\text{Also } f'(x) = 0 \rightarrow e^{-x} = 1$$

$$\therefore x = 0 \text{ and hence there is a minimum turning pt at } (0, 0)$$

$$\therefore e^{-x} - 1 + x \geq 0 \rightarrow e^{-x} \geq 1 - x.$$

c) i)



$$A = 2x \times x + \frac{\pi x^2}{2}$$

$$= 2x^2 + \frac{\pi x^2}{2}$$

$$= \left(2 + \frac{\pi}{2}\right) x^2$$

$$= \left(\frac{4 + \pi}{2}\right) x^2$$

$$\begin{aligned}
\text{cii) } V &= \left(\frac{4 + \pi}{2} \right) \int_0^3 2x^2 dy \\
&= (4 + \pi) \int_0^3 \frac{16 - y^2}{16} dy \\
&= \left(\frac{4 + \pi}{16} \right) \int_0^3 16 - y^2 dy \\
&= \left(\frac{4 + \pi}{16} \right) \left[16y - \frac{y^3}{3} \right]_0^3 \\
&= \left(\frac{4 + \pi}{16} \right) (48 - 9) \\
&= \frac{39}{16} (4 + \pi) \text{ m}^3
\end{aligned}$$

Question 15.

$$\text{a) } \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}.$$

$$\therefore \frac{(s + c)(s^2 - sc + c^2)}{s + c} = \frac{3}{4}$$

$$\therefore 1 - \sin \theta \cos \theta = \frac{3}{4}$$

$$\therefore \sin \theta \cos \theta = \frac{1}{4}$$

$$\therefore \sin 2\theta = \frac{1}{2}$$

$$\therefore 2\theta = n\pi + (-1)^n \left(\frac{\pi}{6} \right)$$

$$\therefore \theta = \frac{n\pi}{2} + (-1)^n \left(\frac{\pi}{12} \right)$$

$$\text{b) i) When } n=2 \text{ } LHS = (1 + x)^2 = 1 + 2x + x^2$$

$$RHS = 1 + 2x$$

$$\text{Since } x^2 > 0, 1 + 2x + x^2 > 1 + 2x$$

Hence true for $n=2$.

Assume true for $n=k, k \geq 2$

$$\text{i.e. } (1 + x)^k > 1 + kx \text{ -----[1]}$$

Prove true for $n=k+1$

$$\text{i.e. } (1 + x)^{k+1} > 1 + (k+1)x$$

$$\text{Now } (1 + x)^{k+1} = (1 + x)(1 + x)^k$$

$$> (1 + x)(1 + kx) \text{ using [1]}$$

$$\text{and since } x > -1 \rightarrow 1 + x > 0$$

$$= 1 + kx + x + kx^2$$

$$> 1 + (k+1)x, \text{ since } kx^2 > 0.$$

Hence by induction is true for all $n \geq 2$.

$$\text{ii) Let } x = -\frac{1}{2n}. \text{ If } n \geq 2, \text{ then this satisfies the}$$

condition of part (i), including $x > -1$.

$$\therefore (1 + x)^n > 1 + nx$$

$$\therefore \left(1 - \frac{1}{2n} \right)^n > 1 - \frac{n}{2n}$$

$$= \frac{1}{2}$$

$$\therefore \left(1 - \frac{1}{2n} \right)^n > \frac{1}{2} \text{ for } n = 2, 3, 4, \dots$$

c) i.e. prove that $a^3 + b^3 - a^2b - ab^2 \geq 0$

$$\begin{aligned}\text{LHS} &= (a+b)(a^2 - ab + b^2) - ab(a+b) \\ &= (a+b)(a^2 - 2ab + b^2) \\ &= (a+b)(a-b)^2 \geq 0, \text{ since } a > 0 \text{ and } b > 0. \\ \therefore a^3 + b^3 &\geq a^2b + ab^2\end{aligned}$$

d) i) By letting $p = q$ in $x + pqy = 5(p+q)$ the equations of the tangents at P and Q are $x + p^2y = 10p$ and $x + q^2y = 10q$ respectively.

ii) By solving simultaneously

$$\begin{aligned}x + p^2y &= 10p \quad \text{---[1]} \\ x + q^2y &= 10q \quad \text{---[2]}\end{aligned}$$

$$\begin{aligned}[1] - [2]: (p^2 - q^2)y &= 10(p - q) \\ (p - q)(p + q)y &= 10(p - q) \\ \therefore y &= \frac{10}{p + q} \quad (p \neq q)\end{aligned}$$

Substituting $y = \frac{10}{p+q}$ in [1] gives

$$\begin{aligned}x + p^2\left(\frac{10}{p+q}\right) &= 10p \\ \therefore x &= 10p - \frac{10p^2}{p+q} \\ &= \frac{10p^2 + 10pq - 10p^2}{p+q} \\ &= \frac{10pq}{p+q}\end{aligned}$$

$\therefore R$ has coordinates $\left(\frac{10pq}{p+q}, \frac{10}{p+q}\right)$.

iii) Since P passes through $(15, 0)$

$$15 + 0 = 5(p+q) \rightarrow p+q = 3$$

$\therefore R$ has coordinates $\left(\frac{10pq}{3}, \frac{10}{3}\right)$.

Hence the locus of R is $y = \frac{10}{3}$. [indep of p and q]

Since $p > 0$ and $q > 0$ then $pq > 0$ and so $x > 0$.

Also since the tangents intersect below the curve:

When $y = \frac{10}{3}$, $x = \frac{15}{2}$, then the locus of R is

$$y = \frac{10}{3}, \text{ for } 0 < x < \frac{15}{2}.$$

Question 16.

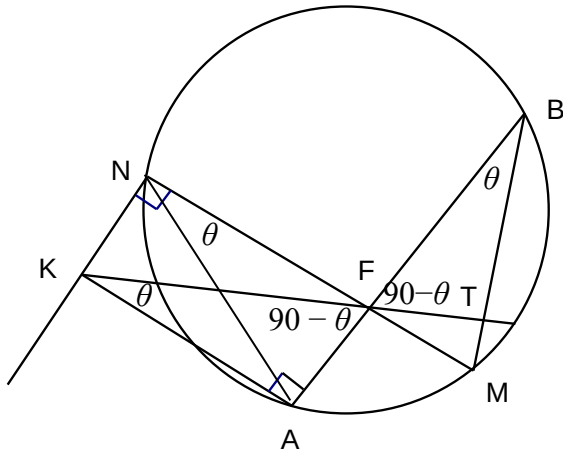
a) $V_s = 2\pi xy \delta x$

$$\begin{aligned}\therefore V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi xy \delta x \\ &= 2\pi \int_0^1 x(\sqrt{x} - x^2) dx \\ &= 2\pi \int_0^1 x^{3/2} - x^3 dx \\ &= 2\pi \left[\frac{2}{5} x^{5/2} - \frac{1}{4} x^4 \right]_0^1 \\ &= 2\pi \left(\left(\frac{2}{5} - \frac{1}{4} \right) - 0 \right) = \frac{3\pi}{10} u^3\end{aligned}$$

b) Let $u = x^n$, $v' = e^{-x}$
 $u' = nx^{n-1}$, $v = -e^{-x}$

$$\begin{aligned}\therefore I_n &= -e^{-x} x^n \Big|_0^1 + \int_0^1 nx^{n-1} e^{-x} dx \\ &= -e^{-1} - 0 + n I_{n-1} \\ &= -\frac{1}{e} + n I_{n-1}\end{aligned}$$

c) i)



Since $\angle KNF + \angle KAF = 180^\circ$

Opposite angles in cyclic quad supplementary
hence $AKNF$ is a cyclic quadrilateral.

ii) Join AN

Let $\angle AKF = \theta$

$\therefore \angle AFK = 90 - \theta$ ($\angle$ sum of ΔAKF)

Also

$\angle ANF = \theta$ ($\angle$ in same segment in cyclic quad $AKNF$)

$\therefore \angle BFT = 90 - \theta$ (vertically opp $\angle AFK$)

and $\angle FBM = \theta$ ($\angle$ in same segment)

$\therefore \angle FTB = 90^\circ$ ($\angle$ sum of ΔFTB)

$\therefore KT$ is perpendicular to MB .

$$d) i) \frac{dQ}{dt} = aQ - bQ^2$$

$$\therefore \frac{dt}{dQ} = \frac{1}{Q(a - bQ)}$$

Using partial fractions let

$$\frac{1}{Q(a - bQ)} = \frac{r}{Q} + \frac{s}{a - bQ}$$

$$\therefore 1 = r(a - bQ) + sQ$$

$$= (s - br)Q + ar$$

$$s - br = 0 \quad \text{and} \quad ar = 1 \rightarrow r = \frac{1}{a} \quad \text{and} \quad s = \frac{b}{a}$$

$$\therefore t = \int \frac{1}{a} \cdot \frac{1}{Q} + \frac{b}{a} \cdot \frac{1}{a - bQ} dQ$$

$$t = \frac{1}{a} \ln Q - \frac{1}{a} \ln(a - bQ) + c$$

$$\text{When } t = 0, Q = Q_0 \Rightarrow c = -\frac{1}{a} \ln \left(\frac{Q_0}{a - bQ_0} \right)$$

$$\therefore t = \frac{1}{a} \ln \left(\frac{Q}{a - bQ} \right) - \frac{1}{a} \ln \left(\frac{Q_0}{a - bQ_0} \right)$$

$$at = \ln \left(\frac{Q}{a - bQ} \times \frac{a - bQ_0}{Q_0} \right)$$

$$\therefore e^{at} = \frac{Q}{a - bQ} \times \frac{a - bQ_0}{Q_0}$$

$$\therefore \frac{Q_0 e^{at}}{a - bQ_0} = \frac{Q}{a - bQ}$$

$$Q = \frac{aQ_0 e^{at}}{(a - bQ_0) + bQ_0 e^{at}} \quad \text{on rearranging}$$

$$ii) \text{ Hence } Q = \frac{aQ_0}{bQ_0 + (a - bQ_0)e^{-at}}$$

$$\text{As } t \rightarrow \infty, e^{-at} \rightarrow 0$$

$$\text{Hence } Q \rightarrow \frac{a}{b} \quad \text{which is constant.}$$

Q13c alternate solution:

Let $z = \cos \theta + i \sin \theta$ and hence $\frac{1}{z} = \cos \theta - i \sin \theta$

$$\therefore \frac{1 + \frac{1}{z}}{1 + z} = \frac{\frac{z + 1}{z}}{1 + z}$$

$$= \frac{1}{z} = \cos \theta - i \sin \theta$$